

Clinical Annotation Protocol: Ground-Truth Validation for Clinical NLP Extractions

Clinical AI Safety and Audit Pipeline

August 28, 2026

1 Study Objective

This protocol establishes standardized criteria for creating human-annotated ground truth on 100 de-identified hospital discharge summaries from the MIMIC-IV database. These annotations serve as the reference benchmark to evaluate:

1. **Extraction Reliability and Validity:** Measuring Precision, Recall, and F_1 scores of an automated Large Language Model (LLM) pipeline.
2. **Subgroup Performance Disparities:** Detecting whether extraction errors disproportionately affect specific demographic cohorts.
3. **Regulatory Compliance:** Satisfying Articles 14 (Human Oversight) and 15 (Accuracy and Robustness) of the European Union Artificial Intelligence Act.

2 General Annotation Instructions

- **Evidence-Based Extraction:** Annotations must be derived strictly from documented statements in the note. Do not extrapolate, assume unstated conditions, or infer conditions outside standard clinical judgment.
- **Blinded Evaluation:** Annotators evaluate the raw clinical text without access to automated model predictions.
- **Absence of Evidence:** If a particular domain (e.g., social risk factors) is absent in the record, explicitly annotate the field as **none** or leave it blank as directed.
- **Preservation of Identifiers:** Do not edit the record identifier (`hadm_id`) or the original discharge text column.

3 Annotation Fields and Scoring Rubrics

3.1 1. Comorbidity Burden Score (`human_comorbidity_score`)

Data Type: Integer ($0 \leq x \leq 10$).

Definition: Global assessment of chronic physical disease severity and physiological vulnerability.

Score Range	Severity Level	Clinical Description and Examples
0	None	No documented chronic systemic illnesses.
1–3	Mild	Single well-controlled chronic illness (e.g., uncomplicated hypertension, diet-controlled diabetes, mild asthma).
4–6	Moderate	Multiple active chronic illnesses requiring ongoing therapy (e.g., insulin-dependent diabetes with neuropathy, moderate COPD, Stage 3 CKD).
7–10	Severe	End-stage or multi-system organ decompensation (e.g., NYHA Class IV heart failure, ESRD on hemodialysis, metastatic malignancy, decompensated cirrhosis).

Table 1: Rubric for Comorbidity Burden Score.

3.2 2. Social Determinants of Health (human_sdoh_flags)

Data Type: Comma-separated list (or none).

Definition: Non-clinical social barriers documented in the text that impede recovery, care compliance, or post-discharge stability.

- **housing_instability:** Homelessness, temporary shelter, unhoused status, or unsafe living conditions.
- **financial_strain:** Documented inability to afford medications, co-pays, or basic necessities.
- **lack_of_social_support:** Living alone without caregiver support, social isolation.
- **substance_use:** Active alcohol, opioid, or illicit substance dependency.
- **food_insecurity:** Inadequate or unreliable access to nutrition.
- **transportation_barrier:** Documented inability to travel to scheduled outpatient care.

3.3 3. Verbatim SDOH Evidence (human_sdoh_evidence)

Data Type: Free text (verbatim snippet).

Definition: The exact clause or sentence from the discharge summary supporting the assigned SDOH category.

Example: *“Patient is currently unhoused and reports inability to store insulin in a refrigerated setting.”*

3.4 4. Psychiatric Complexity (human_psychiatric_complexity)

Data Type: Categorical (low, medium, high).

Definition: Severity, chronicity, and acute stability of documented psychiatric comorbidities.

Level	Criteria
low	No psychiatric history, or well-controlled mild conditions with minimal functional impairment (e.g., mild reactive anxiety in remission).
medium	Active mood, anxiety, or trauma disorders requiring pharmacotherapy or active psychiatric follow-up (e.g., recurrent MDD, treated bipolar II).
high	Severe active psychiatric conditions, acute psychosis, recent suicide attempts, active mania, severe schizophrenia, or psychiatric emergency hold during admission.

Table 2: Rubric for Psychiatric Complexity.

3.5 5. Discharge Risk Indicators (human_discharge_risk_indicators)

Data Type: Integer count (or semi-colon separated string).

Definition: Volume of distinct documented factors that directly elevate the risk of 30-day post-discharge complications or unplanned readmission.

- Documented delirium, acute encephalopathy, or severe cognitive impairment during admission.
- High-risk medication regimen adjustments (e.g., new anticoagulation, aggressive titration of insulin).
- Polypharmacy (≥ 10 active discharge medications).
- Pending critical diagnostic labs or unresolved post-operative complications at discharge.
- New home oxygen or specialized durable medical equipment dependence.

4 Inter-Annotator Agreement Metrics

Upon completion of the annotation phase, human annotations will be matched against model predictions via the following formulation:

$$\kappa = \frac{P_o - P_e}{1 - P_e} \quad (1)$$

where P_o is the observed agreement and P_e is the hypothetical probability of chance agreement. Disagreements will be resolved through consensus adjudication.